Using Force-feedback Control to Reduce Muscle Fatigue during the FES Grasp

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Abstract

FES is broadly used to reduce the need of individuals to rely on assistance from others. However, it is inevitable that the phenomenon of muscle fatigue and the grasped object slips out of hand eventually. In this study, a force feedback system with a FSR glove is adopted not only to prevent slipping of grasped object but also to retard muscle fatigue.

Preliminary clinic trials show that the hemiplegic subject can maintain a constant grasping force in spite of muscle fatigue with the assistance of this system.

1. INTRODUCTION

In the latest decades, many studies of FES are devoted to providing strokes or hemiplegics an alternative way to restore hand function and reduce the need of individuals to rely on assistance from others. Some of them have achieved significant success so that subjects can pick objects up functionally and largely boost their abilities with FES neuroprostheses [1-3]. However, the phenomenon of fatigue, muscle force decreasing over time, is a critical problem frequently encountered in the holding period of FES. If fatigue happens when patients grasping an object such as a cup or a glass, it may slip out of the hand and fall down. In this situation, some studies try to optimize the FES waveforms to retard the phenomenon of fatigue [4]. Others intend to detect the slipping signal from the cutaneous electroneurogram and instantly increase current intensity to hold the object tightly again [5]. In this study, a force feedback system is developed to compensate the force decrease resulting from fatigue and prevent the grasping objects from slipping.

2. METHODS

In our grasping FES system, cup grasping, lateral pinch and three-jaw prehension are provided for subjects' choice and then grasp objects according to their own will. After proper sequence, subjects can grasp objects and hold it in his/her hand, but fatigue will make the grasping force decrease gradually and subjects will eventually release it. By developing a FSR glove which owns 5 FSR sensors, the magnitude of force can be measured. Furthermore, a PI controller is used to adjust current intensities to compensate for force decreasing.

This preliminary study focused on a 42-year-old stroke patient who fails to move hands and fingers voluntarily. The motor status of her upper limbs is Brunnstrom stage II to III. The spasticity is not so prominent to interfere with the hand functioning.

2.1 Stimulated muscles for cup grasping

Contrasting with lateral pinch and three-jaw prehension, cup grasping need largest current intensities and the phenomenon of fatigue is more significant than the other two. For this reason, a force feedback facility for cup grasping is proposed in our study.

In the final stage of cup grasping, all fingers are flexed and thumb is both abducted and flexed. To perform this task, two channels (two stimulation electrodes per channel) stimulate Abductor Pollicis Brevis (AbPB), Flexor Digitorum Sublimes (FDS), Flexor Digitorum Profundus (FDP), Flexor Pollicis Brevis(FPB), Adductor Pollicis(AdP) and Lumbricals simultaneously. Besides, a dual-phased current with frequency of 20 Hz was selected. The electrical pulse width is 300 μs.

2.2 FSR glove

FSR is a polymer thick film device which exhibits a decrease in resistance with an increase in the force applied to the active surface. Combining with a current-to-voltage converter, FSR sensor can easily measure the magnitude of force in terms of voltage. Because of the part-to-part difference, FSR sensors should be calibrated to get better accuracy before use. By means of using standard weights, the relation between force and voltage is thus measured as shown in Fig 1.

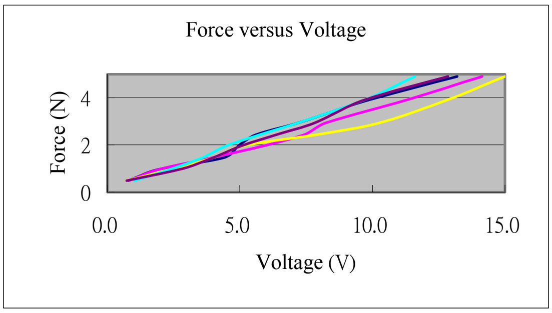

Figure 1 voltage and force relation of each FSR

Previous study shows that FSR is a thin and durable material suitable for sensing grasp force and an epoxy dome is installed on each FSR to measure grasp force effectively [5]. Besides, a thin layer of rubber is stuck to it to increase the friction between FSR and grasping objects. Moreover, for the purpose of donning and doffing easily, all FSR sensors are stuck to a glove in the position that user's hand is used to contacting with objects as shown in Fig 2. Thus, the grasping force can be measured correctly. In this work, a stroke subject wearing the FSR glove is under the clinic trial and the results show that grasp force will decrease gradually as shown in Fig. 3.

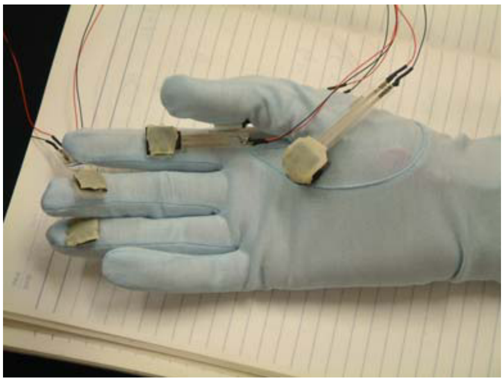

Figure 2 FSR Glove

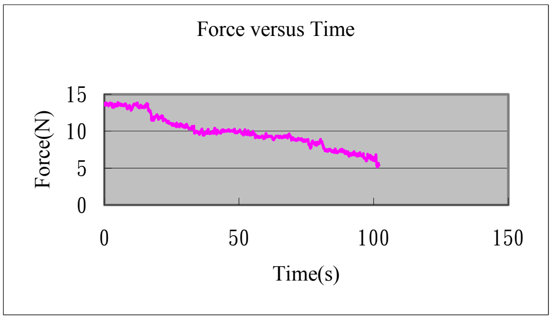

Figure 3 a relation between force and time

2.3 Controller

PI controller is used for compensating the decrease of force. The objective of this controller is to maintain a constant grasping force and parameters of the PI controller are acquired by trials and errors.

3. RESULTS

After trials and errors, K_P=0.2 and K_I=0.1 was chosen. In case that the magnitude of current intensity exceeds the maximum value, the reference input is set to 80 percent of the maximum stimulated force in the whole procedure of grasping. The results show that when a stroke subject grasps an object with FSR glove and PI controller, the current intensities increase to compensate for grasping force as expected. It also shows that the error between reference input and system output is less than 10 percent of the grasp force as shown in Fig. 4 and Fig. 5.

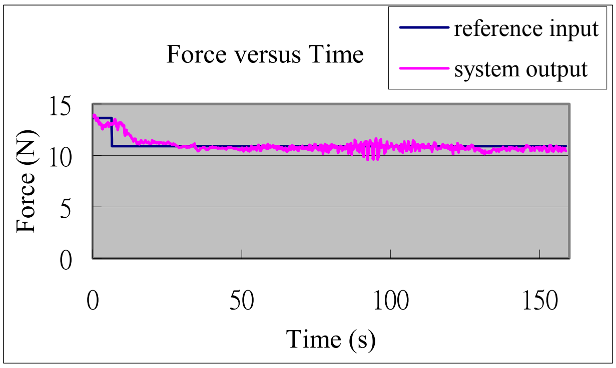

Figure 4 a relation between force and time with force feedback

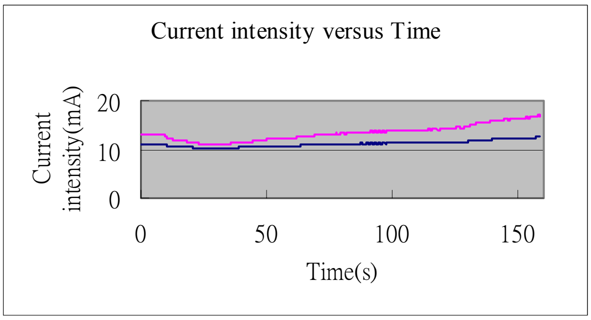

Figure 5 a relation between current intensity and time

4. DISCUSSION AND CONCLUSIONS

The consequence that the force is maintained in the constant value enables subjects to make sure that objects would not slip out. The minimum force for grasping an object is set while subjects were used to setting a much higher level of force to ensure the grasped object would not slip resulting from fatigue.

In spite of the successes of force feedback experiment, the position where FSR is set would influence the measurement if the contact area between hand and grasping object is different from the position expected. In this situation, to optimize numbers of FSR sensors is a further work deserves to investigate.

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